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classdef ProjectileMotionSimulatorAppCode_exported < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        UIFigure                matlab.ui.Figure
        GHTanswer                matlab.ui.control.Label
        SYManswer                matlab.ui.control.Label
        basedonverticaldisplacementyLabel  matlab.ui.control.Label
        SymbolicVerticalVelocityytattimeLabel  matlab.ui.control.Label
        GroundHitTimeLabel      matlab.ui.control.Label
        STIMULATEButton         matlab.ui.control.Button
        Maxheightanswer         matlab.ui.control.Label
        HRanswer                matlab.ui.control.Label
        HorizontalRangeLabel    matlab.ui.control.Label
        MaxHeightLabel          matlab.ui.control.Label
        PROJECTILEMOTIONSIMULATORENG316PROJECTLabel  matlab.ui.control.Label
        ProjectileTypeDropDown  matlab.ui.control.DropDown
        ProjectileTypeDropDownLabel  matlab.ui.control.Label
        InputsPanel             matlab.ui.container.Panel
        GravitationalAccelaration98ms2Label  matlab.ui.control.Label
        InitialHeightEditField  matlab.ui.control.NumericEditField
        InitialHeightEditFieldLabel  matlab.ui.control.Label
        AngleEditField          matlab.ui.control.NumericEditField
        AngleLabel              matlab.ui.control.Label
        InitialVelocityEditField  matlab.ui.control.NumericEditField
        InitialVelocityLabel    matlab.ui.control.Label
        graph2                  matlab.ui.control.UIAxes
        graph3                  matlab.ui.control.UIAxes
        graph1                  matlab.ui.control.UIAxes
    end

    % Callbacks that handle component events
    methods (Access = private)

        % Button pushed function: STIMULATEButton
        function stimulatebutton(app, event)

            v0 = app.InitialVelocityEditField.Value;
            angle = app.AngleEditField.Value;
            Initial_height=app.InitialHeightEditField.Value;
            g =9.81;

            [x,y,v,t]=ProjectileTrajectory(v0,angle,Initial_height,g);
            xlsxwrite('Projectile_Motion_Data',[t', x', y', v'], 'Sheet 1',
'A1:D150');

            for i = 2:length(y)
                if y(i) <= 0
                    ground_hit_time =t(i);
                    app.GHTanswer.Text = sprintf('%.2f s', ground_hit_time);
                    break;
                end
            end
        end
    end
end

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        end

        [MaxHeight, Horizontal_Range]=HeightAndRange(v0, angle,
Initial_height, g, x);
        if MaxHeight < 20
            app.Maxheightanswer.Text = sprintf('Projectile reached low
altitude of %.2f m', MaxHeight);
        elseif MaxHeight < 50
            app.Maxheightanswer.Text = sprintf('Projectile reached
medium altitude of %.2f m', MaxHeight);
        else
            app.Maxheightanswer.Text = sprintf('Projectile reached high
altitude of %.2f m', MaxHeight);
        end

        app.HRanswer.Text = sprintf('%.2f m', Horizontal_Range)
syms T
y_sym = Initial_height + v0 * sind(angle) * T - 0.5 * g * T^2;
vy_sym = diff(y_sym);
time = 1;
Vy_result = double(subs(vy_sym, T, time));
app.SYManwer.Text = sprintf('%.2f m/s', Vy_result);

        PlotGraph(app, x, y, v, t, Initial_height, v0, angle, g)

    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create UIFigure and hide until all components are created
        app.UIFigure = uifigure('Visible', 'off');
        app.UIFigure.Color = [0.8 0.8 0.8];
        app.UIFigure.Position = [100 100 1109 609];
        app.UIFigure.Name = 'MATLAB App';

        % Create graph1
        app.graph1 = uiaxes(app.UIFigure);
        title(app.graph1, 'Title')
        xlabel(app.graph1, 'X')
        ylabel(app.graph1, 'Y')
        zlabel(app.graph1, 'Z')
        app.graph1.Position = [303 293 364 250];

        % Create graph3
        app.graph3 = uiaxes(app.UIFigure);
        title(app.graph3, 'Title')
        xlabel(app.graph3, 'X')
        ylabel(app.graph3, 'Y')
        zlabel(app.graph3, 'Z')

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app.graph3.Position = [438 33 474 242];

% Create graph2
app.graph2 = uiaxes(app.UIFigure);
title(app.graph2, 'Title')
xlabel(app.graph2, 'X')
ylabel(app.graph2, 'Y')
zlabel(app.graph2, 'Z')
app.graph2.Position = [697 295 401 248];

% Create InputsPanel
app.InputsPanel = uipanel(app.UIFigure);
app.InputsPanel.Title = 'Inputs';
app.InputsPanel.Position = [19 402 240 151];

% Create InitialVelocityLabel
app.InitialVelocityLabel = uilabel(app.InputsPanel);
app.InitialVelocityLabel.HorizontalAlignment = 'right';
app.InitialVelocityLabel.Position = [-2 100 81 22];
app.InitialVelocityLabel.Text = 'Initial Velocity:~';

% Create InitialVelocityEditField
app.InitialVelocityEditField = uieditfield(app.InputsPanel,
'numeric');
app.InitialVelocityEditField.HorizontalAlignment = 'left';
app.InitialVelocityEditField.Position = [93 100 100 22];

% Create AngleLabel
app.AngleLabel = uilabel(app.InputsPanel);
app.AngleLabel.Position = [7 69 46 22];
app.AngleLabel.Text = 'Angle:~';

% Create AngleEditField
app.AngleEditField = uieditfield(app.InputsPanel, 'numeric');
app.AngleEditField.HorizontalAlignment = 'left';
app.AngleEditField.Position = [90 69 103 22];

% Create InitialHeightEditFieldLabel
app.InitialHeightEditFieldLabel = uilabel(app.InputsPanel);
app.InitialHeightEditFieldLabel.Position = [7 39 74 22];
app.InitialHeightEditFieldLabel.Text = 'Initial Height:~';

% Create InitialHeightEditField
app.InitialHeightEditField = uieditfield(app.InputsPanel,
'numeric');
app.InitialHeightEditField.HorizontalAlignment = 'left';
app.InitialHeightEditField.Position = [90 39 103 22];

% Create GravitationalAccelaration98ms2Label
app.GravitationalAccelaration98ms2Label =
uilabel(app.InputsPanel);
app.GravitationalAccelaration98ms2Label.HorizontalAlignment =
'center';
app.GravitationalAccelaration98ms2Label.FontSize = 10;

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app.GravitationalAccelaration98ms2Label.FontAngle = 'italic';
app.GravitationalAccelaration98ms2Label.FontColor = [1 0 0];
app.GravitationalAccelaration98ms2Label.Position = [7 18 208 22];
app.GravitationalAccelaration98ms2Label.Text = 'Gravitational
Acceleration = 9.8m/s^2';

% Create ProjectileTypeDropDownLabel
app.ProjectileTypeDropDownLabel = uilabel(app.UIFigure);
app.ProjectileTypeDropDownLabel.HorizontalAlignment = 'right';
app.ProjectileTypeDropDownLabel.FontSize = 14;
app.ProjectileTypeDropDownLabel.Position = [13 360 97 22];
app.ProjectileTypeDropDownLabel.Text = 'Projectile Type';

% Create ProjectileTypeDropDown
app.ProjectileTypeDropDown = uidropdown(app.UIFigure);
app.ProjectileTypeDropDown.Items = {'-select-', 'Rock', 'Ball'};
app.ProjectileTypeDropDown.FontSize = 14;
app.ProjectileTypeDropDown.Position = [131 360 100 22];
app.ProjectileTypeDropDown.Value = '-select-';

% Create PROJECTILEMOTIONSIMULATORENG316PROJECTLabel
app.PROJECTILEMOTIONSIMULATORENG316PROJECTLabel =
uilabel(app.UIFigure);
app.PROJECTILEMOTIONSIMULATORENG316PROJECTLabel.BackgroundColor
= [0.6353 0.0784 0.1843];

app.PROJECTILEMOTIONSIMULATORENG316PROJECTLabel.HorizontalAlignment =
'center';
app.PROJECTILEMOTIONSIMULATORENG316PROJECTLabel.FontName =
'Arial Rounded MT Bold';
app.PROJECTILEMOTIONSIMULATORENG316PROJECTLabel.FontSize = 18;
app.PROJECTILEMOTIONSIMULATORENG316PROJECTLabel.FontWeight =
'bold';
app.PROJECTILEMOTIONSIMULATORENG316PROJECTLabel.Position = [183
565 729 23];
app.PROJECTILEMOTIONSIMULATORENG316PROJECTLabel.Text =
'PROJECTILE MOTION SIMULATOR- ENG316 PROJECT';

% Create MaxHeightLabel
app.MaxHeightLabel = uilabel(app.UIFigure);
app.MaxHeightLabel.FontSize = 14;
app.MaxHeightLabel.Position = [17 295 85 16];
app.MaxHeightLabel.Text = 'Max Height: ';

% Create HorizontalRangeLabel
app.HorizontalRangeLabel = uilabel(app.UIFigure);
app.HorizontalRangeLabel.FontSize = 14;
app.HorizontalRangeLabel.Position = [19 328 131 22];
app.HorizontalRangeLabel.Text = 'Horizontal Range: ';

% Create HRanswer
app.HRanswer = uilabel(app.UIFigure);
app.HRanswer.FontSize = 14;
app.HRanswer.Position = [141 329 224 22];

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app.HRanswer.Text = '';

% Create Maxheightanswer
app.Maxheightanswer = uilabel(app.UIFigure);
app.Maxheightanswer.WordWrap = 'on';
app.Maxheightanswer.FontSize = 14;
app.Maxheightanswer.Position = [101 274 250 43];
app.Maxheightanswer.Text = '';

% Create STIMULATEButton
app.STIMULATEButton = uibutton(app.UIFigure, 'push');
app.STIMULATEButton.ButtonPushedFcn = createCallbackFcn(app,
@stimulatebutton, true);
app.STIMULATEButton.Position = [73 51 158 54];
app.STIMULATEButton.Text = 'STIMULATE';

% Create GroundHitTimeLabel
app.GroundHitTimeLabel = uilabel(app.UIFigure);
app.GroundHitTimeLabel.FontSize = 14;
app.GroundHitTimeLabel.Position = [17 245 111 22];
app.GroundHitTimeLabel.Text = 'Ground Hit Time: ';

% Create SymbolicVerticalVelocityyttattimelsLabel
app.SymbolicVerticalVelocityyttattimelsLabel =
uilabel(app.UIFigure);
app.SymbolicVerticalVelocityyttattimelsLabel.FontSize = 14;
app.SymbolicVerticalVelocityyttattimelsLabel.Position = [13 209
324 25];
app.SymbolicVerticalVelocityyttattimelsLabel.Text = 'Symbolic
Vertical Velocity y'(t) at time 1s ';

% Create basedonverticaldisplacementyLabel
app.basedonverticaldisplacementyLabel = uilabel(app.UIFigure);
app.basedonverticaldisplacementyLabel.FontSize = 14;
app.basedonverticaldisplacementyLabel.Position = [13 185 276 25];
app.basedonverticaldisplacementyLabel.Text = 'based on vertical
displacement y():';

% Create SYManswer
app.SYManswer = uilabel(app.UIFigure);
app.SYManswer.FontSize = 14;
app.SYManswer.Position = [241 186 79 22];
app.SYManswer.Text = '';

% Create GHTanswer
app.GHTanswer = uilabel(app.UIFigure);
app.GHTanswer.FontSize = 14;
app.GHTanswer.Position = [131 245 225 22];
app.GHTanswer.Text = '';

% Show the figure after all components are created
app.UIFigure.Visible = 'on';
end
end

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```
% App creation and deletion
methods (Access = public)

    % Construct app
    function app = ProjectileMotionSimulatorAppCode_exported

        % Create UIFigure and components
        createComponents(app)

        % Register the app with App Designer
        registerApp(app, app.UIFigure)

        if nargin == 0
            clear app
        end
    end

    % Code that executes before app deletion
    function delete(app)

        % Delete UIFigure when app is deleted
        delete(app.UIFigure)
    end
end
end
```

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